Estimating Persistent Explanatory Variables: A Machine Learning Approach

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Paper: SSRN 4636846 Code and example data

R&D and patent production

R&D investment is a central input into innovation.

- Firms invest in research over several years before producing patents.
- R&D changes slowly within firms. Much of its variation reflects persistent differences across firms.
- Firm fixed effects control for those differences, but leave only within-firm changes to estimate the R&D–patent relation.

How much information remains when the explanatory variable is highly persistent?

Paper, Introduction. Detailed R&D results follow the methods and simulations.

Persistent regressors and firm fixed effects

- Firm fixed effects control for unobserved, time-invariant firm characteristics.
- Many economic variables change slowly within firms: R&D, ownership, organizational practices, and managerial characteristics.
- FE estimates rely on the variation remaining within firms. That variation can be small and noisy.
- The resulting trade-off is between controlling for persistent heterogeneity and retaining information for precise estimation.

Question: Can we select informative firm indicators rather than include all of them?

Paper, Introduction and Section 2. Persistence alone does not imply FE inconsistency.

What changes when we include firm effects?

$$z_{it} = \eta_i + \nu_{it}, \qquad \tilde{z}_{it} = z_{it} - \bar{z}_i = \nu_{it} - \bar{\nu}_i.$$

- Demeaning removes the permanent firm component η_i.
- A larger between-firm share leaves less within-firm variation.
- Over a short observation window, stronger persistence can further reduce variation after demeaning.
- Under strict exogeneity and identification, FE can remain consistent while estimates become imprecise.
- Measurement error is a separate problem: removing the true signal can increase the relative importance of remaining noise.

Paper, Sections 2 and 4. The decomposition above is illustrative.

Selective estimation and empirical evidence

1. Setting and methods
   Select firm indicators using PRL and DML in linear and Poisson models.

2. Simulations
   Examine between-firm variation, serial persistence, and the structure of firm components.

3. Diagnostics
   Describe the available variation and calibrate error and power targets.

4. Empirical applications
   Return to R&D and innovation, then examine managerial ownership and firm value.

Panel regression framework

$$y_{i,t+1} = \beta_0 + \gamma z_{it} + \mathbf{x}_{it}^{\top}\boldsymbol{\beta} + \alpha_i + \varepsilon_{i,t+1}.$$

- z_{it}: persistent explanatory variable of interest.
- $\mathbf{x}_{it}$: observed controls, including year indicators.
- α_i: time-invariant firm heterogeneity.
- γ: coefficient of interest.
- Empirical inference uses standard errors clustered by firm.

In the ownership application, ownership and its square enter jointly.

Paper, Section 2.1. Empirical outcomes and controls follow the paper's timing conventions.

Conventional estimators

Estimator	Variation used	Main consideration
Pooled OLS	Between and within firms	Persistent omitted firm factors can confound the coefficient.
Firm FE	Deviations from firm means	Controls unrestricted firm intercepts; needs usable within variation.
FD	Consecutive-year changes	Removes firm intercepts; small annual changes can be especially noisy.

- FE and FD remove the same time-invariant firm component.
- Their relative efficiency also depends on the error process.
- Comparing coefficients requires attention to assumptions, samples, and the scale of the regressor change.

Paper, Section 2.1; Wooldridge (2002); Breuer and deHaan (2024).

Selective firm effects

$$y_{i,t+1} = \beta_0 + \gamma z_{it} + \mathbf{x}_{it}^{\top}\boldsymbol{\beta} + \boxed{\sum_{s \in \mathcal{S}} \alpha_s \text{dummy}_{s,i}} + \varepsilon_{i,t+1}.$$

- Pooled estimation includes no firm indicators.
- Full FE allows unrestricted firm intercepts, subject to a normalization.
- PRL and DML select a subset of candidate indicators.
- In the applications, year indicators are also candidates. The reported controls enter as variables of interest.

Paper, Section 2.2. Identification and nuisance-approximation conditions remain necessary.

Economic intuition for selective indicators

- After observed characteristics and common time effects are controlled for, some firms may retain large persistent deviations.
- Distinctive technologies, organizational practices, or governance arrangements can motivate such departures.
- A sparse approximation represents the firm component with a common baseline and a limited set of substantial deviations.
- Selection is a statistical adjustment. It does not identify a firm's culture or prove that unselected firm effects are exactly zero.

Paper, Introduction, Section 2, and Internet Appendix concentration analysis. The selected share is sample- and specification-dependent.

Post-regularization LASSO (PRL)

1. Outcome equation
   Select firm/year indicators by LASSO; refit on the selected indicators and obtain outcome residuals.

2. Regressor equations
   Repeat selection and post-LASSO fitting for R&D and each included control, obtaining residualized regressors.

3. Target coefficient
   Regress the residualized outcome on the residualized regressors; use firm-clustered inference.

The main linear application uses BIC for LASSO tuning.

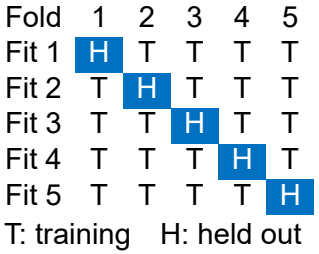
Paper, Section 2.2; Chernozhukov, Hansen, and Spindler (2015). Implementation: `poregress`.

Double machine learning (DML)

1. Split firm-year observations into five folds.
2. Fit the nuisance models, including selection and post-LASSO refitting, on four training folds.
3. Use those fitted models to obtain residuals for the held-out fold.
4. Repeat until every observation has out-of-fold residuals.
5. Pool those residuals to estimate the target coefficient.

Cross-fitting separates nuisance fitting from evaluation.
Each training fit uses fewer observations, but the final stage uses residuals from all folds.

Paper, Section 2.3; Chernozhukov et al. (2018). Implementation: `xporegress`, five folds.



Count-based outcomes

$$\mu_{it} = \exp\left(\beta_0 + \gamma z_{it} + \mathbf{x}_{it}^{\top}\boldsymbol{\beta} + \sum_{s \in \mathcal{S}} \alpha_s \text{dummy}_{s,i} \right).$$

Here μ_{it} is the conditional mean of the count outcome.

- Poisson and FE Poisson use no firm indicators and the full set, respectively.
- PRL Poisson selects indicators using Poisson LASSO and weighted linear auxiliary regressions.
- The target parameters solve residual-based moment equations.
- DML Poisson adds cross-fitting.

Paper, Section 2.4; Belloni, Chernozhukov, and Wei (2016). Patent and citation counts; adjusted citations are excluded.

Simulation design

The count outcome follows a negative binomial distribution:

$$\mathbb{E}[y_{it} \mid z_{it}, x_{it}] = \exp(0.2z_{it} + x_{it}).$$

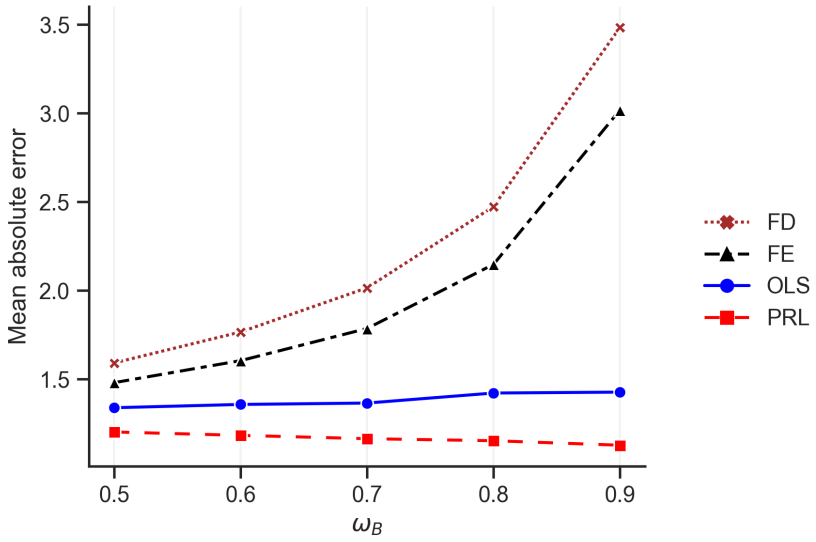
- $N = 1{,}000$, $T = 10$, dispersion parameter 0.5; 10,000 replications per cell.
- Four designs: between-firm share, serial persistence, prevalence of a firm component, and a component shared with an observed control.
- Linear comparisons: OLS, FE, FD, PRL. Count comparisons: Poisson, FE Poisson, PRL Poisson.
- MSE and MAE use the paper's common reference $\gamma = 0.2$; one-sided power is $\Pr(t > 1.645)$.
- DML is omitted from simulations because of computational cost.

Paper, Section 3. Linear estimators use the same count outcomes; the exponential-mean coefficient is their common comparison benchmark, not an assumed correctly specified linear slope.

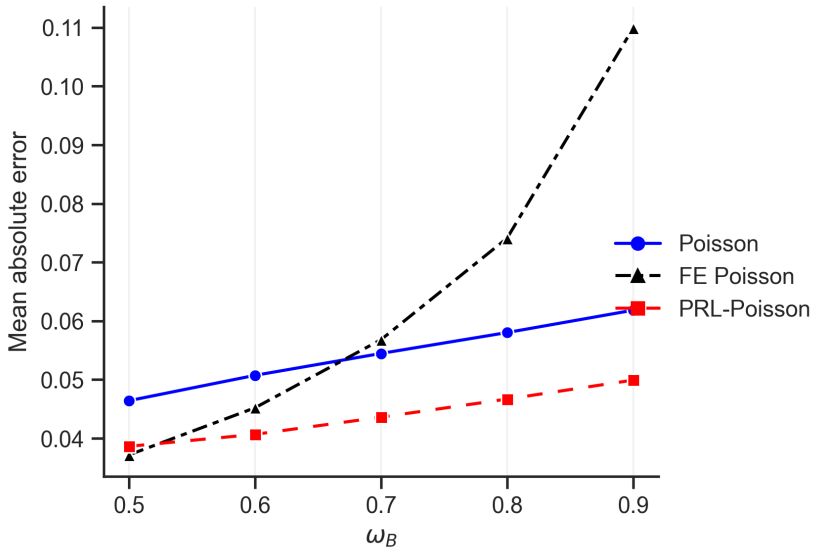
Between-firm variation

$z_{it} = 0.5\eta_i + 0.5\nu_{it}$; horizontal axis: ω_B.

Linear: MAE



Poisson: MAE



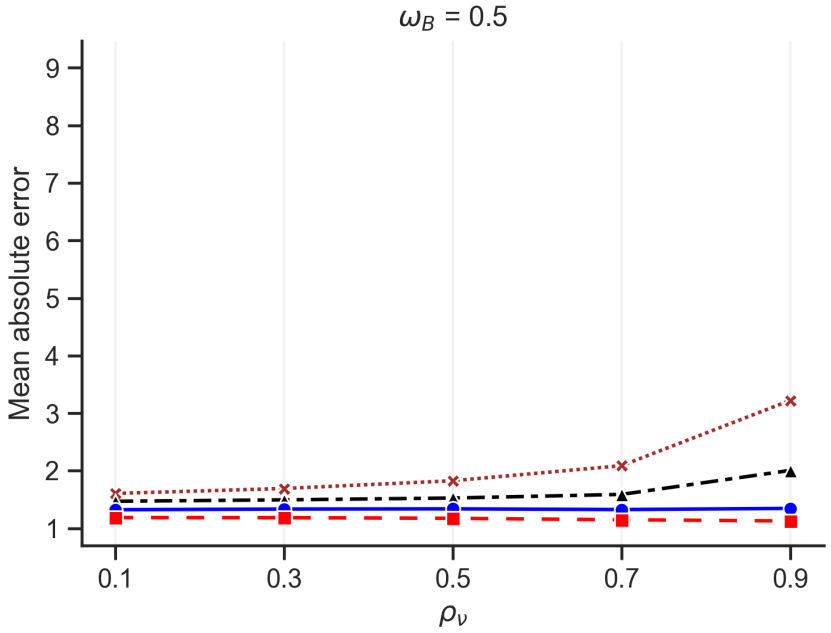
- Linear PRL has the lowest error in these cells.
- FE Poisson performs well at moderate between-firm shares; its error rises when the share becomes high.

Paper, Figure 1, MAE panels. $N = 1,000$, $T = 10$, 10,000 replications.

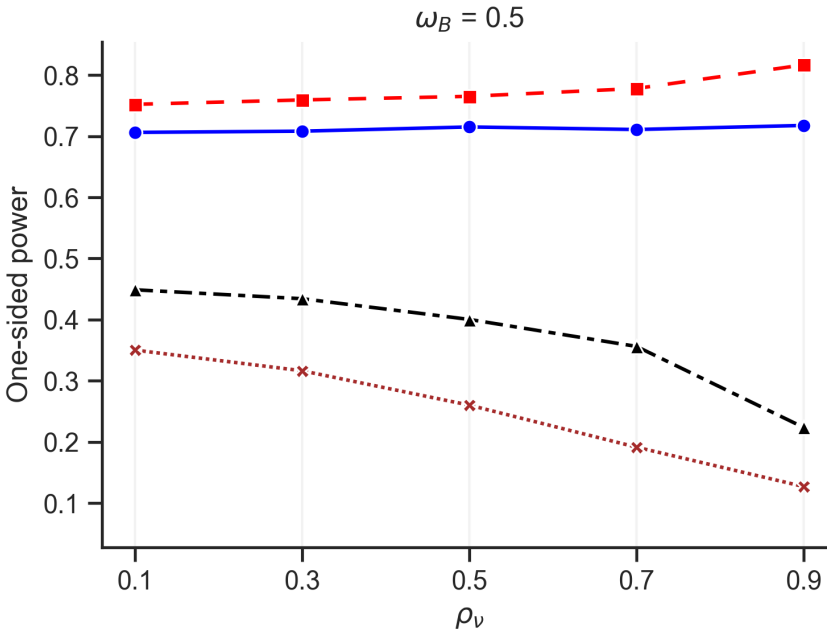
Serial persistence

$$\nu_{it} = \rho_\nu \nu_{i,t-1} + \sigma_\nu \sqrt{1-\rho_\nu^2}\, \varepsilon_{\nu,it}.$$

Mean absolute error



One-sided power



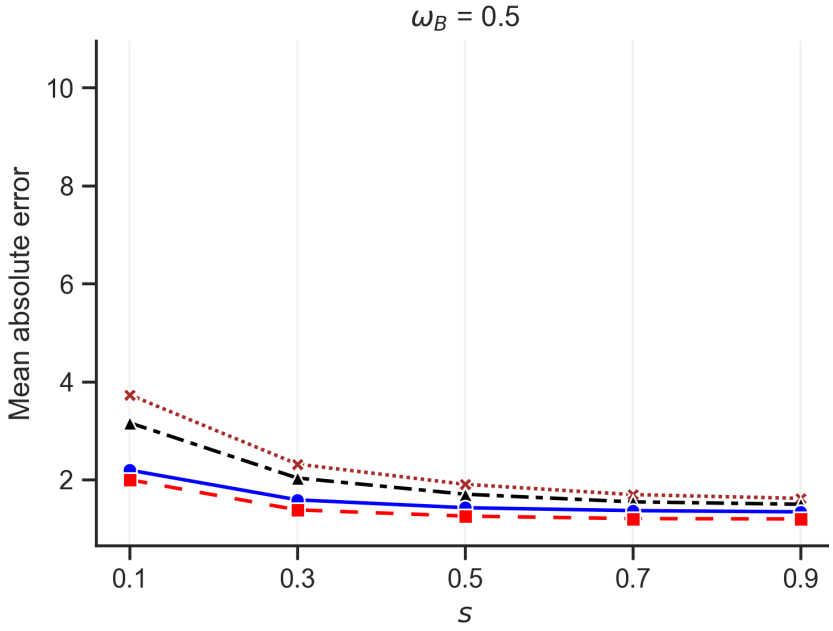
OLS: blue solid; PRL: red dashed; FE: black dash-dot; FD: brown dotted.

FE and FD errors increase and power declines as persistence rises, particularly when the between-firm share is high.

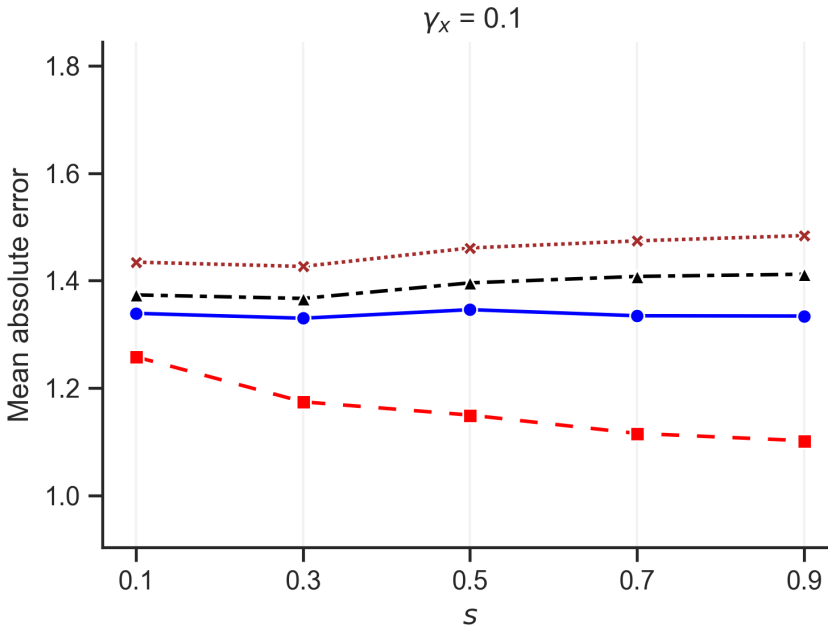
Paper, Figure 2. Enlarged $\omega_B = 0.5$ cells; the paper also reports 0.7 and 0.9.

Prevalence and shared firm components

Prevalence: linear MAE



Shared component: linear MAE



OLS: blue solid; PRL: red dashed; FE: black dash-dot; FD: brown dotted.

PRL performs well in the linear comparisons. Poisson rankings also depend on the design and error criterion.

Paper, Figures 3 and 4. Enlarged MAE cells: $\omega_B = 0.5$ for prevalence and $\gamma_x = 0.1$ for the shared component. Backup slides show additional parameter cells.

Variation and persistence diagnostics

- **D1: between-firm variance share**
  How much of the regressor's variation lies between firms?
  $\widehat{\omega}_B = \dfrac{\sum_i T_i(\bar{z}_i - \bar{z})^2}{\sum_{it}(z_{it} - \bar{z})^2}$

- **D2: within-firm persistence**
  $\widehat{\rho}_i = \text{Corr}_t(z_{it}, z_{i,t-1})$ Median and interquartile range of firm-level autocorrelations.

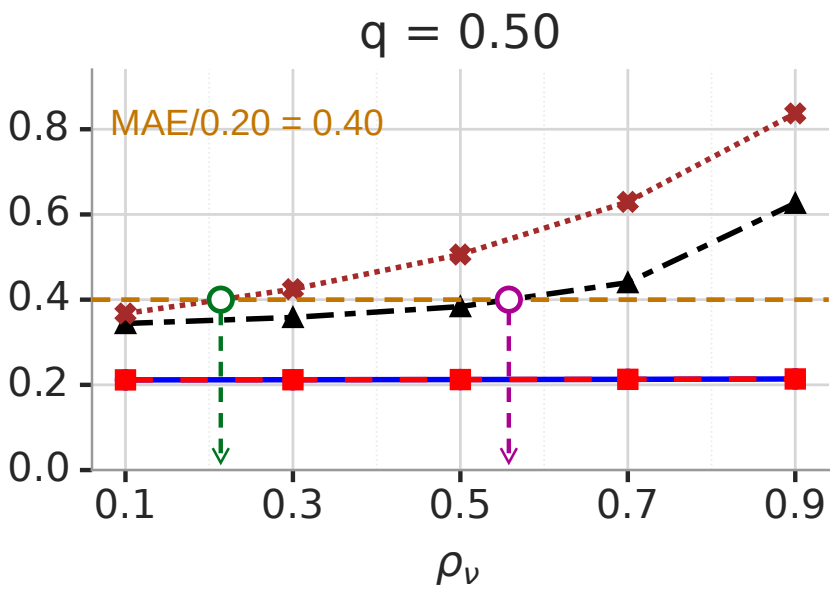
- **D3: fixed-effects absorption**
  A_z is the auxiliary-regression R^2 of z_{it} The fraction of regressor variation explained by the specified fixed effects.

- **Between/within correlation-R^2 ratio**
  How closely predictions track firm means versus within-firm deviations. $R^2_{\text{B,corr}} = \text{Corr}_i(\bar{y}_i, \bar{m}_i)^2$,
  $R^2_{\text{W,corr}} = \text{Corr}_{it}(y_{i,t+1} - \bar{y}_i, \widehat{m}_{i,t+1} - \bar{m}_i)^2$.

Paper, Section 4.1. These diagnostics describe variation. They do not establish exogeneity or supply universal estimator cutoffs.

Diagnostic guidance under an assumed DGP

$z = z^* + m$, $y = \alpha_i + 0.2z^* + \varepsilon$, $\text{Var}(z^*) = 1$.
Calibration: $\sigma_\varepsilon = 3$, $\omega_B = 0.5$, $q = \sigma_m^2/\sigma_v^2 = 0.5$.



OLS: blue; PRL: red; FE: black; FD: brown.

Target: MAE/0.2 < 0.4

- FD crosses at $\rho_v = 0.214$.
- FE crosses at $\rho_v = 0.558$.

The diagnostic values at each crossing provide observable benchmarks under this calibration.

For power ≥ 0.9: FD fails throughout the plotted range. FE crosses near $\rho_v = 0.591$ in the separate power guide.

Paper, Section 4.3 and Figure 5; IA power guide. Crossings are interpolated and scenario-specific.

From persistence to observable diagnostics

At each MAE/$0.2 = 0.4$ crossing, read the diagnostic values from the same simulation.

Estimator	Persistence	D1	D2	Between/within R^2
FD	0.214	0.480	−0.010	13.436
FE	0.558	0.542	0.132	19.981

- D1, D2, and the between/within R^2 ratio translate the persistence crossing into observable diagnostic benchmarks.
- D3 equals D1 with firm effects alone. With year or other effects, evaluate absorption for that exact fixed-effect structure.
- These benchmarks apply to this calibration; they are not universal estimator cutoffs.

Paper, Section 4.3 and Figure 5: $q = 0.50$, $\sigma_\varepsilon = 3$, $(\sigma_\eta^2, \sigma_\nu^2) = (0.5, 0.5)$. Persistence denotes ρ_ν. Values are interpolated and rounded to three decimals. The figure does not provide a separate D3 cutoff with additional fixed effects.

R&D and innovation data

- U.S. public firms from CRSP/Compustat; patent and citation information from PatentsView.
- Exclude financial and utility firms; require nonmissing, nonnegative total assets.
- Full sample: 86,341 firm-years and 11,544 firms, 1975–2000.
- Patenting-firm subsample: 45,913 firm-years and 4,312 firms.
- Explanatory variables precede the innovation outcome by one year.

Paper, Section 5.2.1. FE and FD use smaller estimation samples, reported on the result slides.

R&D input and innovation outcomes

Main measure: five-year accumulated R&D divided by total assets.
Robustness: R&D/ME and $\ln(1 + \text{R\&D})$, using the same five-year accumulation; ME is market equity.

- Linear outcomes: $\ln(1 + \text{Patents})$, $\ln(1 + \text{Citations})$, and $\ln(1 + \text{Adjusted citations})$.
- Poisson outcomes: patent and citation counts.
- Controls include size, age, capital intensity, missing R&D, Tobin's Q, profitability, leverage, cash, financing constraints, institutional ownership, and competition.
- Use the paper's winsorized variables and full control set.

Paper, Section 5.2.1 and Tables 2 and 4. Alternative R&D measures include R&D/ME and $\ln(1 + \text{R\&D})$.

R&D variation and persistence

Diagnostic	R&D/Assets
D1: between-firm variance share	0.801
D2: median within-firm autocorrelation	0.851
D2: interquartile range	0.625–0.950
D3: absorption by firm and year effects	0.806
Patent model: between correlation R^2	0.310
Patent model: within correlation R^2	0.020
Between/within correlation-R^2 ratio	15.207

Most variation is between firms, and R&D is highly persistent within firms. These statistics describe the available signal; they do not establish exogeneity.

Paper, Table 1, full pooled sample. The ratio uses saved precision, not the displayed rounded R^2 values.

R&D and patents: linear estimates

Dependent variable: $\ln(1 + \text{Patents})$

	OLS	FE	FD	PRL	DML	adjHT
R&D/Assets	.593***	.041	−.031	.199***	.213***	.220***
	(.042)	(.028)	(.023)	(.018)	(.014)	(.027)
Selected firms				1,215	1,711	
Firm selection				10.52%	14.82%	
Observations	86,341	84,620	73,319	86,341	86,341	86,341

- PRL and DML retain positive, significant associations.
- FE and FD coefficients are close to zero and statistically insignificant.
- Coefficient differences alone cannot identify their source.

Paper, Table 2, Panel A. Full controls; firm-clustered SE in parentheses. FE removes 1,721 singleton observations. FD uses consecutive-year differences. * $p < .10$, ** $p < .05$, *** $p < .01$.

Citations and adjusted citations

	OLS	FE	FD	PRL	DML	adjHT
ln(1+Citations)						
R&D/Assets	1.396***	−.051	−.176**	1.397***	1.364***	.551***
	(.084)	(.068)	(.080)	(.083)	(.050)	(.065)
ln(1+Adjusted citations)						
R&D/Assets	.590***	.033	−.051*	.210***	.198***	.231***
	(.045)	(.031)	(.031)	(.019)	(.015)	(.030)
Observations	86,341	84,620	73,319	86,341	86,341	86,341

The positive relation under pooled and selective estimation also appears for citation-based outcomes. FE coefficients remain insignificant; FD estimates are negative.

Paper, Table 2, Panels B and C. Full controls and firm-clustered SE. Stars use the same thresholds as the preceding slide.

R&D and innovation: Poisson estimates

	Poisson	FE Poisson	PRL Poisson	DML Poisson
Patent counts				
R&D/Assets	2.305***	−.248	2.407***	2.312***
	(.187)	(.255)	(.161)	(.120)
Firm selection			10.33%	21.29%
Citation counts				
R&D/Assets	2.094***	−.125	2.299***	2.262***
	(.219)	(.252)	(.256)	(.113)
Firm selection			10.32%	20.79%
Observations	86,341	84,620	86,341	86,341

Pooled and selective estimates imply similar positive associations. The full-FE estimates are insignificant.

Paper, Table 4. Full controls; firm-clustered SE. Firm selection percentages exclude year indicators. *** $p < .01$.

Managerial ownership and firm value

- ExecuComp and Compustat: 23,879 firm-years and 2,158 firms.
- Source years: 2006–2024. Lagged-variable level-estimation sample: 2007–2024.
- Ownership aggregates all named executives' directly held common shares and unexercised exercisable options, divided by shares outstanding.
- Dependent variable: Tobin's Q, constructed following Himmelberg et al. (1999).
- Ownership, ownership squared, and controls enter with a one-year lag.

Paper, Section 5.3.1 and Table 5. Ownership and its square are not winsorized.

Ownership variation and persistence

Diagnostic	Ownership
D1: between-firm variance share	0.700
D2: median within-firm autocorrelation	0.694
D2: interquartile range	0.382–0.866
D3: absorption by firm and year effects	0.726
Tobin's *Q* model: between correlation R^2	0.230
Tobin's *Q* model: within correlation R^2	0.051
Between/within correlation-R^2 ratio	4.469

Ownership is persistent, with much of its variation between firms. This reproduces the central information trade-off in a different economic setting.

Paper, Table 1. Diagnostics use the pooled application sample. Ratios use saved precision.

Ownership and firm value: estimates

Dependent variable: Tobin's Q

	OLS	FE	FD	PRL	DML	adjHT
Ownership	1.945***	−.312	1.393***	2.034***	1.395***	1.778**
	(.600)	(.757)	(.528)	(.717)	(.383)	(.707)
Ownership^2	−1.521*	−1.377	−2.422***	−3.610***	−2.851***	−2.699***
	(.827)	(.991)	(.683)	(1.174)	(.695)	(.957)
Observations	23,879	23,766	21,516	23,879	23,879	23,879

- OLS, FD, PRL, and DML show a positive level term and negative curvature.
- Both FE terms are individually insignificant.
- Marginal effects depend on the initial ownership level.

Paper, Table 5. Firm-clustered SE; full controls. FE excludes 113 singletons. * $p < .10$, ** $p < .05$, *** $p < .01$.

Selected firm indicators

Selected firm indicators	PRL	DML
Linear patents	1,215 (10.52%)	1,711 (14.82%)
Linear citations	500 (4.33%)	922 (7.99%)
Linear adjusted citations	1,168 (10.12%)	1,856 (16.08%)
Poisson patents	1,192 (10.33%)	2,458 (21.29%)
Poisson citations	1,191 (10.32%)	2,400 (20.79%)
Managerial ownership	580 (26.88%)	725 (33.60%)

- Candidate firms: 11,544 in R&D; 2,158 in ownership.
- There is no universal “10–20%” selected share.
- Selected indicators may capture outcome or regressor nuisance variation.

Paper, Tables 2, 4, and 5. Counts refer to firms only; year indicators are reported separately in the paper.

Robustness and interpretation

R&D application

- Alternative innovation outcomes and R&D measures.
- Patenting-firm sample and observations with reported R&D.
- Alternative tuning choices and cross-fitting folds.

Ownership application

- Top-five-executive ownership and alternative sample periods.
- Results vary across periods; historical-sample curvature is not uniformly concave.

These checks assess sensitivity. They do not establish a unique causal explanation for differences across estimators.

Paper, Sections 5.2–5.3 and the corresponding Internet Appendix robustness tables.

Conclusions

- Persistent regressors create a difficult balance between controlling heterogeneity and preserving useful variation.
- PRL and DML provide selective specifications for linear and Poisson applications.
- Simulations identify settings where PRL improves performance, while also showing that no estimator dominates uniformly.
- The two applications yield positive R&D–innovation associations and concave ownership–value relations under selective estimation.
- Interpretation requires attention to identifying assumptions, measurement quality, and available within-firm information.

Paper, Conclusion. Empirical coefficient differences alone do not establish causality or a unique explanation for the differences.

Thank you!

Questions and comments

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Code and example data

Paper: SSRN 4636846
Replication code, example data, and sparse Stata package

Estimating Persistent Explanatory Variables:
A Machine Learning Approach

Adjusted Hausman–Taylor

- The adjusted HT estimator uses internal instruments and GMM moments involving demeaned regressors.
- Its validity requires strict exogeneity and additional restrictions on controls used as instruments in levels.
- R&D coefficients: patents 0.220, citations 0.551, and adjusted citations 0.231; all significant at 1%.
- Ownership: level 1.778 and square -2.699, significant at 5% and 1%, respectively.
- We treat it as an assumption-dependent supplementary comparison.

Paper, Section 6 and Tables 2 and 5; Hausman and Taylor (1981).

Stata implementation

The linear application uses BIC and five cross-fitting folds:

```
poregress y z x, controls(i.firmID i.year) ///
    vce(cluster firmID) selection(bic, gridminok) ///
    rseed(1103)

xporegress y z x, controls(i.firmID i.year) ///
    vce(cluster firmID) selection(bic, gridminok) ///
    xfolds(5) rseed(1103)
```

- `x` stands for the full control-variable list.
- Poisson counterparts: `popoisson` and `xpopoisson`.
- The sparse package supplies `sparse_poreg` and `sparse_xporeg` for large *linear* models.

Commands adapted from `RDPatent/po_xpo_adjHT.do`. Replication code, example data, and sparse Stata package.

Practical reporting guidance

1. Report between-firm variation, within-firm persistence, FE absorption, and between/within correlation R^2.
2. Explain the assumptions supporting pooled, FE, FD, and selective specifications.
3. Report estimation observations, singleton removal, and the annual-pair sample for FD.
4. Report selected firm/year counts and sensitivity to tuning and folds.
5. Use a substantively justified DGP when translating diagnostics into error and power targets.

Paper, Section 4. Diagnostics characterize the available variation and do not independently validate identification.

Between- and within-firm R-squared

Use predictions from observed regressors and year effects:

$$\widehat{m}_{i,t+1} = \widehat{\beta}_0 + \widehat{\gamma} z_{it} + \mathbf{x}_{it}^{\top} \widehat{\boldsymbol{\beta}}.$$

These predictions exclude the estimated firm indicators.

$$R^2_{\text{B,corr}} = \text{Corr}_i(\bar{y}_i, \bar{m}_i)^2,$$

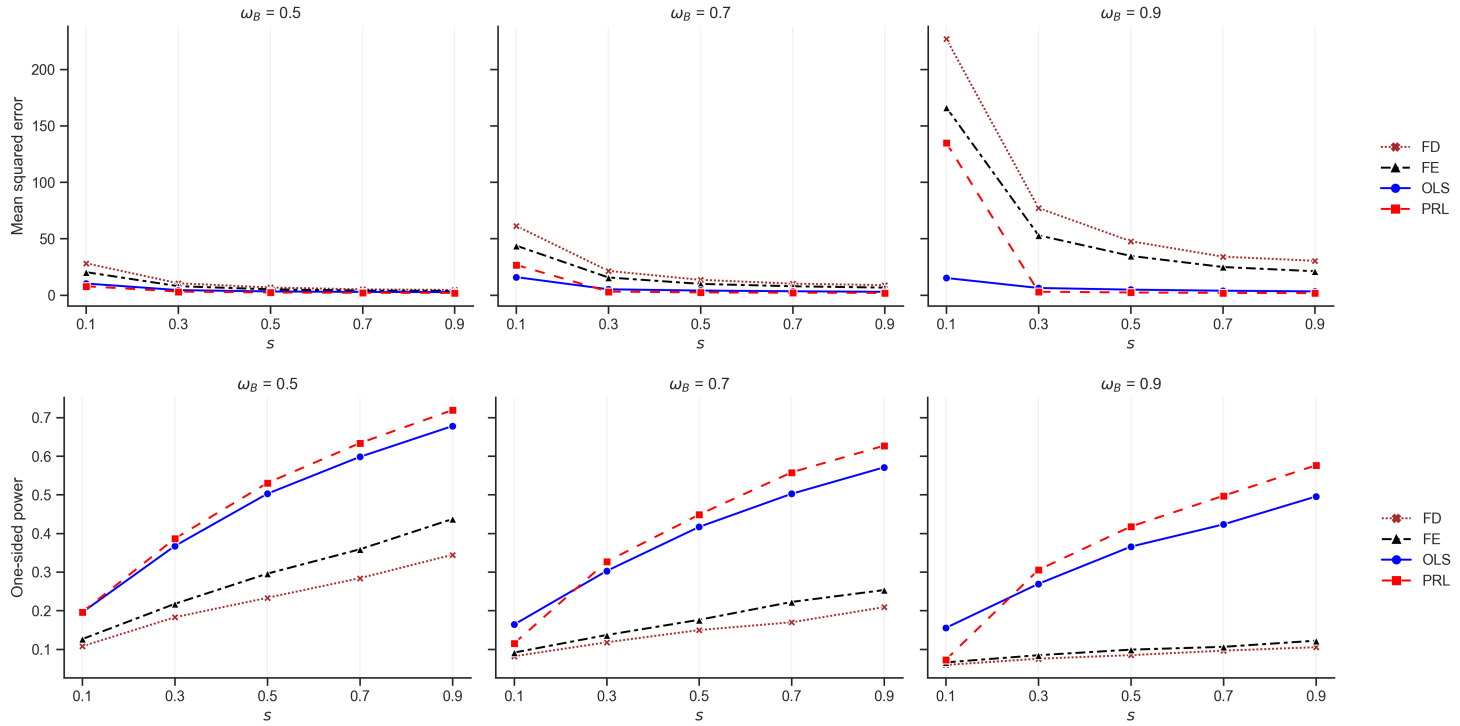
$$R^2_{\text{W,corr}} = \text{Corr}_{it}(y_{i,t+1} - \bar{y}_i, \widehat{m}_{i,t+1} - \bar{m}_i)^2.$$

- The between measure compares firm-average outcomes and predictions.
- The within measure compares deviations from firm means.
- These differ from full-model R^2 including fitted firm effects, and generally depend on the fitted slopes.

Paper, Section 4.1. These squared correlations are not additive variance shares.

Prevalence of the time-invariant component

$z_{it} = 0.5 D_i \eta_i + 0.5 \nu_{it}$, $D_i \sim$ Bernoulli(s).

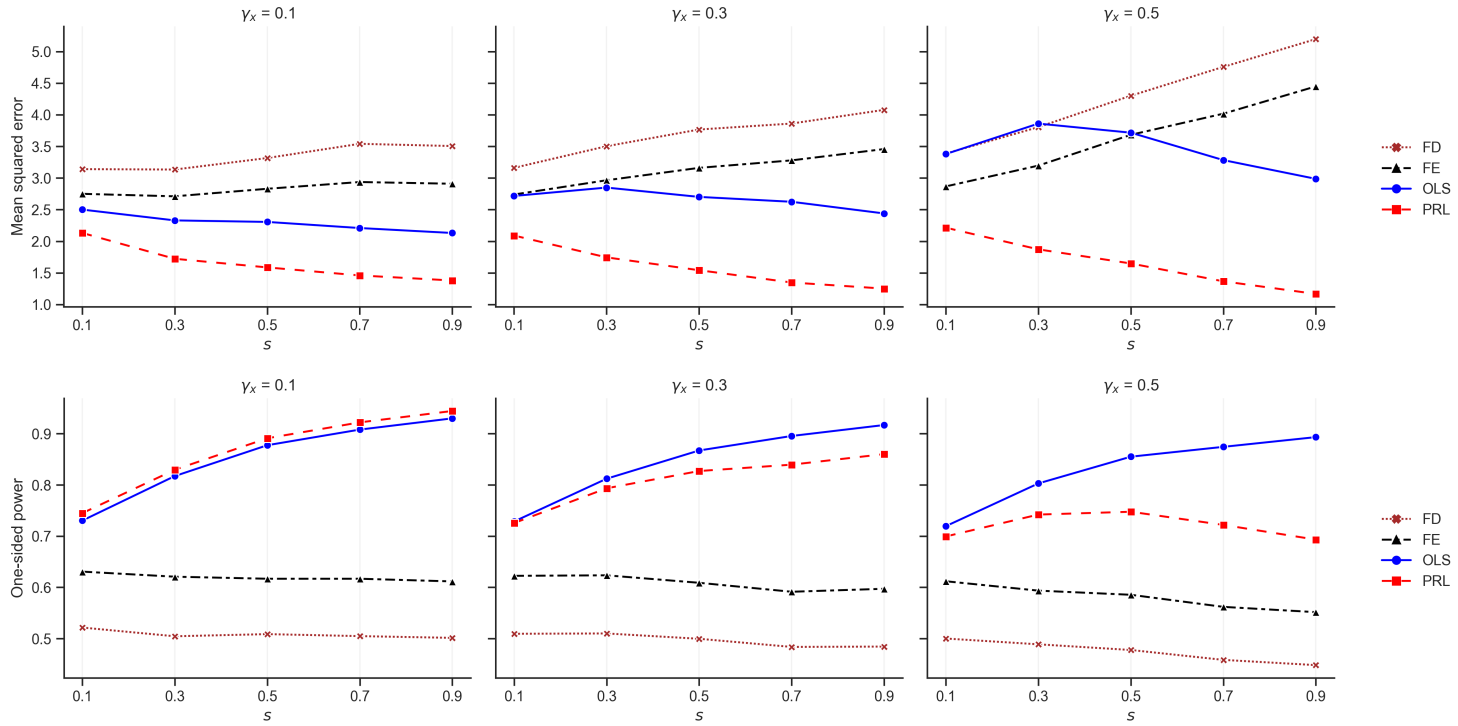


OLS: blue solid; PRL: red dashed; FE: black dash-dot; FD: brown dotted.

Paper, Figure 3. Rows: MSE and power. Columns: $\omega_B = .5, .7, .9$. The scale $\sigma_\nu^2 = s\sigma_\eta^2(1-\omega_B)/\omega_B$ holds the population share fixed as s varies.

Shared firm component

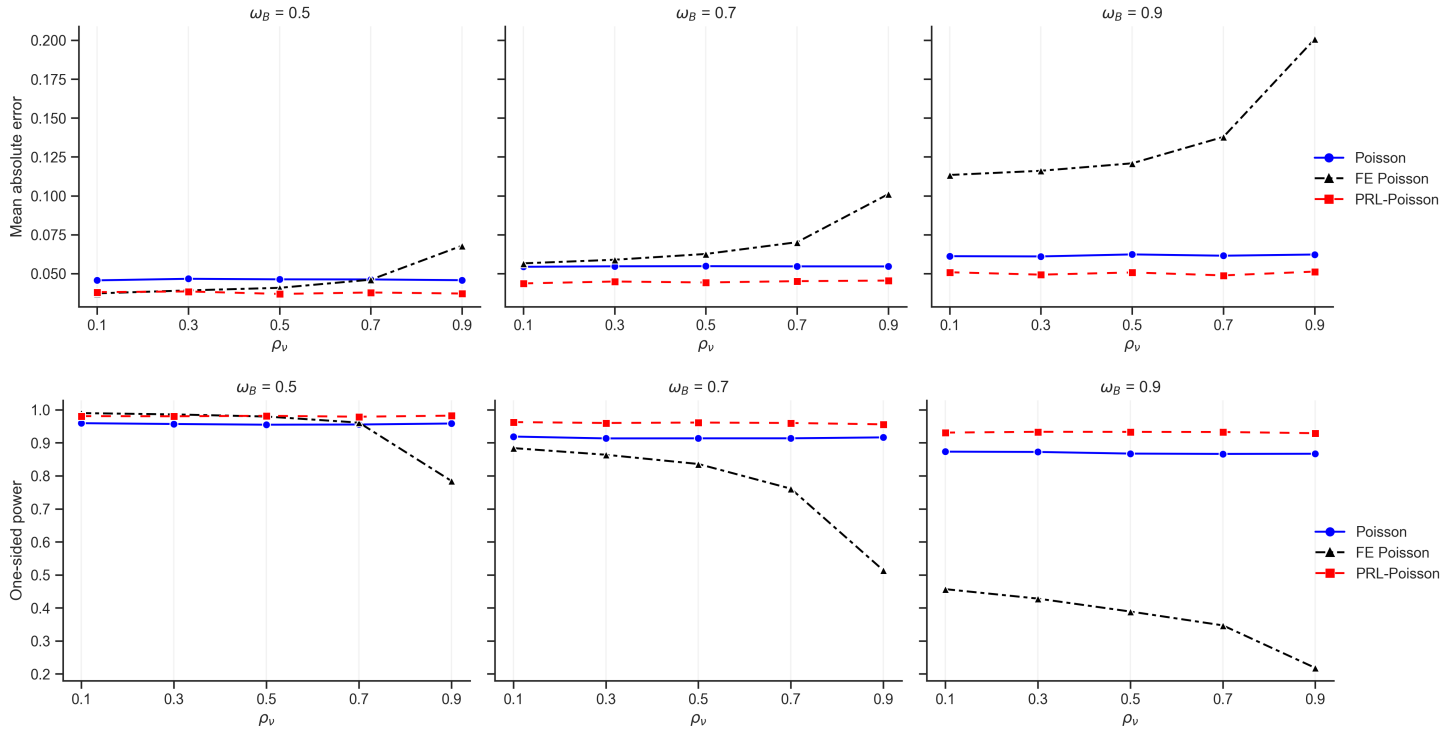
$$z_{it} = D_i\eta_i + \nu_{it},\ x_{it} = \gamma_x D_i\eta_i + \kappa_{it}.$$



OLS: blue solid; PRL: red dashed; FE: black dash-dot; FD: brown dotted.

Paper, Figure 4. Rows: MSE and power. Columns: $\gamma_x = .1, .3, .5$. Baseline variance calibration .7; the population between-firm variance share is $0.7s/(0.7s + 0.3)$.

Serial persistence: Poisson comparisons



FE Poisson can perform well with moderate between-firm variation and low persistence. Its performance deteriorates in more persistent settings; PRL Poisson does not dominate every cell.

Internet Appendix, count-outcome serial-persistence figure. Rows: MAE and power. Columns: $\omega_B = .5, .7, .9$; 10,000 replications.

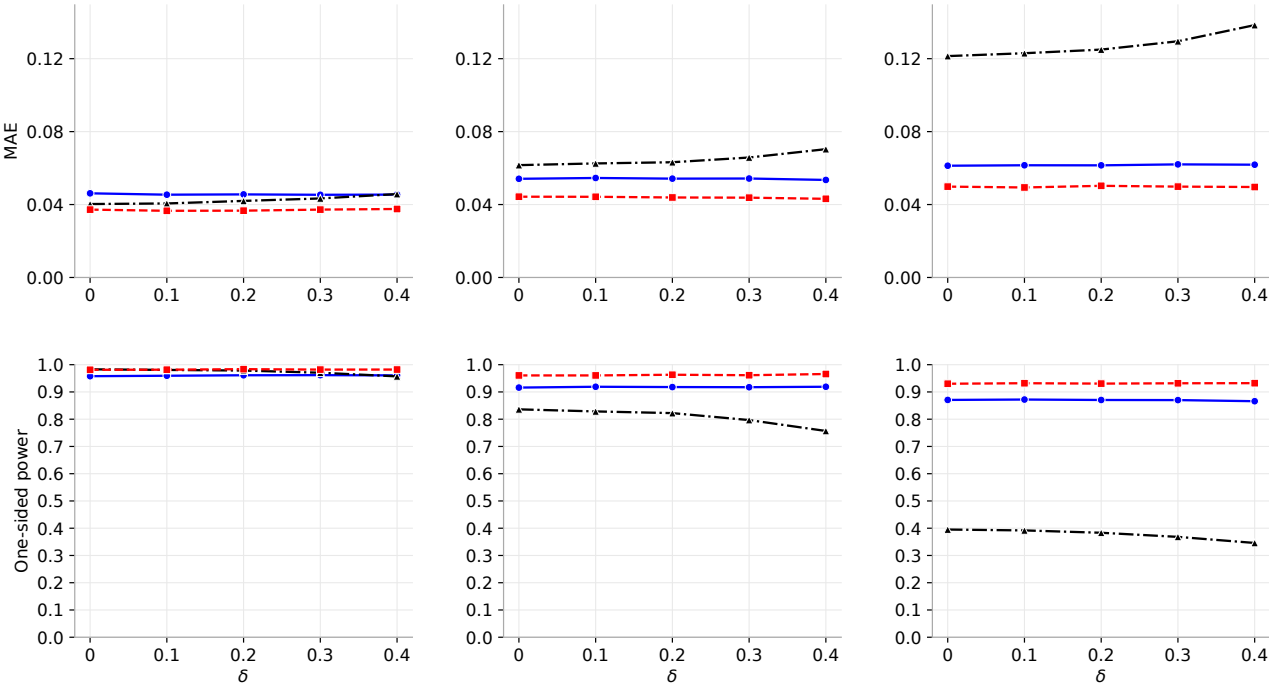
Larger panels and continuous outcomes

- Larger panels: $N = 5{,}000$, $T = 10$, and 2,000 replications per parameter combination.
- Continuous outcome: a nonnegative lognormal–chi-squared mixture with conditional mean $\exp(0.2z_{it} + x_{it})$.
- The Internet Appendix repeats the between-firm, serial-persistence, prevalence, and shared-component designs.
- Larger panels generally reduce errors and increase power, but relative rankings depend on the design and loss criterion.
- The sparse linear PRL package reduces memory and computational costs in large panels.

Paper, Section 3 and Internet Appendix simulation extensions. These simulation comparisons omit DML.

Heterogeneous persistence across firms

Half the firms have $\rho_L = 0.5 - \delta$ and half $\rho_H = 0.5 + \delta$.
Columns: $\omega_B = 0.5, 0.7, 0.9$. Rows: MAE and one-sided power.



Poisson: blue solid; FE Poisson: black dash-dot; PRL Poisson: red dashed.

Internet Appendix, count outcome, $N = 1,000$, $T = 10$, 10,000 replications. $\delta = 0, .1, .2, .3, .4$. PRL Poisson has the lowest MAE throughout.

R&D robustness: patenting firms

Dependent variable: $\ln(1 + \text{Patents})$

	OLS	FE	FD	PRL	DML	adjHT
R&D/Assets	.448***	.035	−.003	.406***	.399***	.161***
	(.045)	(.031)	(.028)	(.044)	(.025)	(.030)
Selected firms				128	255	
Firm selection				2.97%	5.91%	
Observations	45,913	45,679	40,944	45,913	45,913	45,913

- Restrict to firms with at least one patent during the sample period.
- The positive pooled and selective associations remain.
- Selected proportions are lower than in the full sample.

Internet Appendix, table labelled `tab:rd_linear_patenting`, Panel A. Full controls; firm-clustered SE. FE excludes 234 singletons. *** $p < .01$.

Ownership robustness: top five executives

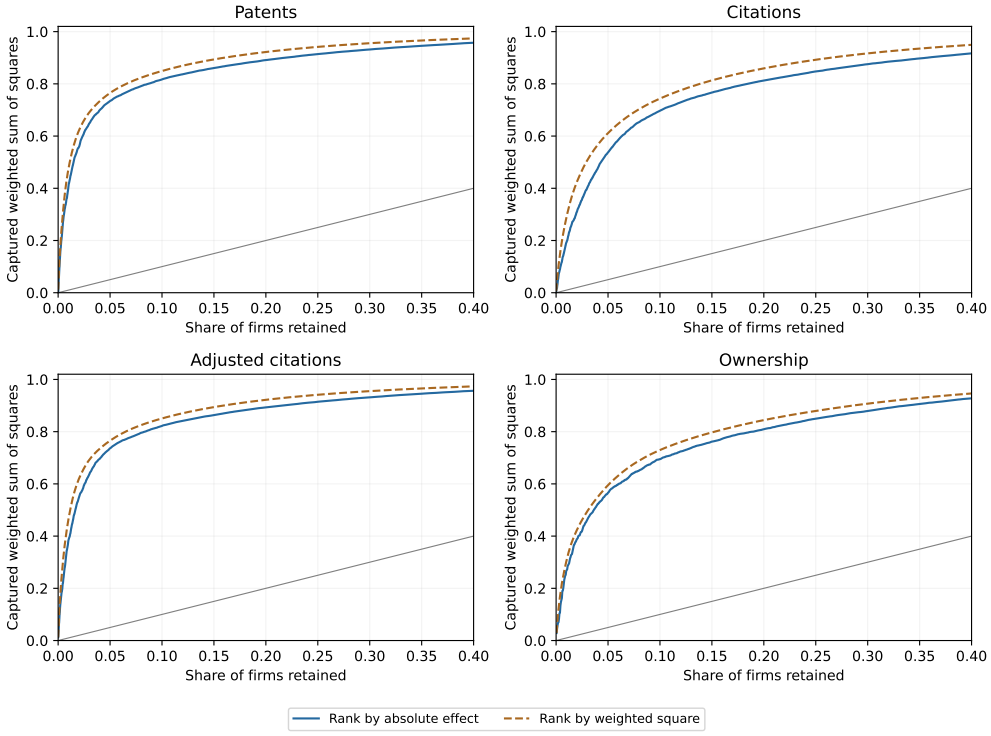
Dependent variable: Tobin's Q; level sample 2007–2024.

	OLS	FE	FD	PRL	DML	adjHT
Ownership	2.066***	−.035	1.244**	1.305**	1.201***	2.017***
	(.608)	(.754)	(.503)	(.524)	(.402)	(.711)
Ownership^2	−1.612*	−1.399	−2.183***	−2.326***	−2.357***	−2.704***
	(.832)	(.991)	(.687)	(.860)	(.733)	(.964)
Observations	23,892	23,779	21,522	23,892	23,892	23,892

The positive level and negative squared terms remain under pooled and selective estimation. FE terms remain individually insignificant.

Internet Appendix, `ownership_top5_main_period_table.tex`. Scope-specific filters yield a slightly different sample from the all-named-executive baseline. Firm-clustered SE; conventional significance stars.

Concentration of estimated firm effects



A limited subset can capture a large share of the fitted firm-effect component in some applications. This is descriptive evidence, affected by estimation noise.

Internet Appendix concentration analysis. Solid: rank by absolute centered effect; dashed: rank by observation-weighted squared effect. These are unrestricted-FE rankings, not actual PRL/DML selected sets.

Implementation and selected references

- Chernozhukov, Hansen, and Spindler (2015): post-regularization inference.
- Chernozhukov et al. (2018): double/debiased machine learning.
- Belloni, Chernozhukov, and Wei (2016): post-selection inference for generalized linear models.
- Breuer and deHaan (2024): use and interpretation of fixed effects.
- Jennings et al. (2024): measurement error and fixed effects.
- Cohn, Liu, and Wardlaw (2022): count-data regression in finance.

Code, data, logs, and the sparse linear package
Replication code, example data, and sparse Stata package

Paper: SSRN 4636846

Full bibliographic details are in the paper.

Selected versus non-selected firms

Firm-level comparisons using the actual linear PRL selections.

Characteristic	Selected mean	Non-selected mean
R&D–patents: 1,215 selected; 10,329 non-selected		
R&D/Assets	0.259	0.121
Log market equity	12.530	10.525
Tobin's *Q*	3.144	1.795
Cash/assets	0.301	0.148
Ownership: 580 selected; 1,578 non-selected		
Managerial ownership	0.086	0.038
Log sales	6.624	7.478
Capital/sales	0.888	0.232
EBITDA/sales	0.083	0.143

Selected firms are larger in the R&D application but smaller in the ownership application.

Paper, table labelled `tab:selected_firms`, selected rows. Average each variable within firm, then average firm means within each group, giving firms equal weight. These are unconditional descriptive comparisons. Ownership and ratios are fractions. PRL selections are specific to each application.

Top-ranked PRL-selected firms

Five largest signed firm-indicator coefficients in each post-selection refit.

Rank	R&D–patents	Ownership–firm value
1	Lucent Technologies	NVIDIA
2	IBM	Palantir Technologies
3	Aptiv	Datadog
4	General Electric	CAVA Group
5	AT&T Inc.	AppFolio

- Hold the saved PRL firm and year selections fixed and refit the unpenalized outcome regression with the original controls.
- Rank selected firms by their signed fitted intercept deviations, from largest to smallest. Non-selected firms form the reference group.

Internet Appendix, tables labelled `tab:rd_selected_top50` and `tab:ownership_selected_top50`. Names follow the appendix's Compustat metadata. These are conditional fitted-effect rankings, not rankings by selection frequency or absolute effect size.

PRL selection robustness

Re-estimate PRL with BIC in 50 whole-firm draws at each fraction, retaining all usable observations of each sampled firm.

Application	Firms sampled	Selected	Retention	Jaccard
R&D–patents	90%	11.1%	96.1%	0.878
	80%	11.6%	97.7%	0.842
Ownership	90%	27.2%	96.7%	0.880
	80%	28.5%	97.5%	0.859

- R&D coefficient medians: 0.199 and 0.189, versus 0.199 in the full sample.
- Ownership coefficient medians: 1.253 and 0.866, versus 2.034 in the full sample. The squared-term medians remain negative.

Internet Appendix, `tab:selection-prl-subsamples`. Cells are medians across 50 draws. Selected: share of sampled firms selected. Retention: share of baseline-selected firms selected again, conditional on being sampled. Jaccard: intersection/union of selected sets within sampled firms. Draws are without replacement, with 80% nested in 90%. These checks assess subsample sensitivity under BIC, not stability across tuning rules or DML partitions.